Symbols and Conventions

An unadorned observation is measured; a hat marks a prediction or estimate. A tilde marks a constructed surrogate or adjusted quantity. The subscript old means evaluation at a fixed behavior-policy snapshot. A bar over critic parameters marks a fixed target copy. Bold **a** always means an action chunk, while a means one action, which can itself be a vector. Bold attention and mechanics symbols distinguish matrices or vectors from scalar learning quantities.

Interaction, context, and time

Symbol	Meaning
t	Integer control-step index. A local imagined rollout starts at $t = 0$.
$t_{\text{phys},t},\ \Delta t$	Physical timestamp of step t and duration of one control interval. These have units of time.
$s_t,\ o_t,\ \widehat{o}_t$	Full physical state, measured observation, and predicted observation.
$h_t,\ \widehat{h}_t$	Available observation–action history; an imagined history starts with real evidence and then appends generated transitions.
$a_t,\ \mathbf{a}_t$	One action and a proposed chunk of consecutive actions. Replanning may replace the unexecuted part of a chunk.
$H,\ d_a,\ K$	Number of actions in a chunk, dimension of one action, and number executed before replanning; $K \leq H$.
H_{roll}	Number of transitions in an imagined rollout, distinct from chunk length.
T_{ep}	Terminal step index of an episode, not a duration in seconds.
$g,\ C$	Task specification and optional demonstration context. Context need not include action labels.
$c_t = (h_t, g, C)$	Policy conditioning information. Write c for a generic training example.
C_{dyn}	Additional evidence about physical conditions, such as earlier interactions.
L	Number of recent observation latents supplied to a predictor.
$o_{j:k},\ z_{j:k}$	Sequence from index j through index k, inclusive.
$i,\ N$	Training-example index and number of examples in a batch.
$m,\ M$	Noise/time-sample index and number of samples in a Monte Carlo loss estimate.
$j,\ k$	Local indices: j indexes a sampling grid or candidates; k indexes a future offset. The use is stated locally.

Models and their parameters

Symbol	Meaning
$\pi_\theta,\ \theta$	Policy distribution and its trainable parameters. Its argument determines whether the distribution is over an action or a whole chunk.
θ_{old}	Fixed parameters of the behavior policy that collected the current batch.
v_θ	Policy velocity network used for flow sampling and flow-loss evaluation. It is not the robot's physical velocity.
$p_\phi,\ \phi$	Conditional world-model distribution and its parameters.
P_ϕ	Point-prediction alternative for latent dynamics, rather than a distribution.
$R_\psi,\ \psi$	Learned reward model and its parameters.
$V_\nu,\ \nu,\ \bar{\nu}$	Learned critic, its trainable parameters, and a fixed parameter copy used to form targets.
$q_\zeta,\ \zeta,\ \widehat{y}$	Outcome evaluator, its parameters, and its predicted label. The label is not automatically a calibrated probability.
$E_\alpha,\ D_\beta$	Observation encoder and decoder, with parameters α and β.
z_t	Observation latent $E_\alpha(o_t)$.
$E_{\text{tr}},\ z^\star$	Fixed transition encoder and its future-derived latent target. The star marks a target here, not an optimal action.
$f_\omega,\ \omega,\ \widetilde{z}$	Action adapter, its parameters, and the latent condition predicted before the future is observed.
η_{learn}	Positive learning rate in a gradient-descent update.
$\Pr, \text{sign}$	Probability of an event; sign of a scalar (negative, zero, positive).
$\mathcal{D}$	Dataset of aligned training examples.
$d_{\text{image}},\ d_{\text{latent}}$	Chosen reconstruction and latent-prediction losses. Neither is inherently a physical distance.
$\mathbf{Q}, \mathbf{K}, \mathbf{V}$	Attention query, key, and value matrices. These differ from action values Q^π, execution prefix K, and critic values V^π.
d_{key}	Shared dimension of an attention query and key.

Rewards, returns, and policy updates

Symbol	Meaning
$r_t,\ \widehat{r}_t$	Reward for the transition beginning at t, and its model prediction.
γ	Discount factor per stated decision interval, between zero and one.
G_t	Discounted return from step t through the episode.
G_i^{target}	Fixed return target for critic training sample i.
$\widehat{G}_t^{(\lambda)}$	Bootstrapped imagined return target using return-mixing parameter λ.
$V^\pi,\ Q^\pi$	Expected return under policy π; expected return after a specified first action and then following π.
$\mathcal{A}^\pi,\ \widehat{\mathcal{A}}$	True advantage $Q^\pi - V^\pi$ and an estimated advantage.

Symbol	Meaning
$d_t,\ m_t = 1 - d_t$	Terminal indicator and continuation mask. A computational cutoff alone is not termination.
δ_t	One-step temporal-difference residual.
λ	Return/advantage mixing parameter in $[0, 1]$; never log signal-to-noise ratio in this book.
b_{adv}	Advantage-derived bin or indicator used as a policy input.
$J(p),\ p$	Expected return and left-action probability in the two-action teaching example.
$f(a)$	Fixed outcome score in the score-function gradient derivation.
$S,\ S_{\text{PPO}}$	Local empirical policy surrogate and PPO's clipped surrogate, both maximized.
$\rho,\ \tilde{\rho}$	Exact current-to-behavior action-density ratio and FPO's flow-loss surrogate ratio. They are not interchangeable.
ϵ_{clip}	Positive clipping width in the policy surrogate; distinct from Gaussian noise ϵ.
$\mathcal{L},\ \ell_\theta$	A minimized loss and a loss for one sampled noise/time pair. Subscripts identify the objective or parameters.
$\widehat{\mathcal{L}}_{\theta,i},\ \widehat{\mathcal{L}}_{\text{old},i}$	Monte Carlo flow-loss estimates for sample i, under current and fixed old policies.
$\mathcal{V}_{\text{RISE}},\ \widehat{\mathcal{A}}_{\text{RISE}}$	RISE-specific progress/value predictor and chunk score. The latter is not assumed equal to every textbook advantage estimator.
$u_{\text{hall}},\ \chi,\ \widetilde{r}$	Estimated hallucination severity, penalty strength, and adjusted reward.
$\mathcal{U}$	Fixed set of candidate actions for a decision comparison.
$\widehat{Q},\ Q_{\text{ref}}$	Model-based decision score and independently estimated reference outcome value under a stated continuation rule.
$a^\star,\ a^\star_{\text{ref}}$	Candidates maximizing the model score and reference value, respectively.

Generative sampling and likelihood bounds

Symbol	Meaning
$\tau,\ X_\tau$	Artificial flow time from noise (0) to action data (1), and the intermediate action sample.
$\epsilon,\ I$	Standard Gaussian noise and identity covariance of the appropriate dimension.
u_τ	Known reference velocity along a constructed training path.
τ_j	Point on the numerical sampling grid.
σ	Forward noising coordinate from data to noise in Appendix A; not a standard deviation.
$\ell_{\text{snr}}(\sigma)$	Log signal-to-noise ratio along that forward process.
$w,\ \widehat{\epsilon}_\theta$	Nonnegative loss-weighting function and model-predicted noise.

Symbol	Meaning
$\mathcal{L}_\theta^w$	Weighted denoising loss under the stated schedule and parameterization.
$B_\theta,\ \mathcal{K}_\theta$	Variational lower bound on log likelihood and its nonnegative gap, for a compatible probabilistic construction.
b_{const}	A term independent of policy parameters, common to old and current losses in the cancellation argument.

Physical interfaces and computational cost

Symbol	Meaning
$\mathbf{q}_t$	Robot configuration used to determine a coordinate transformation.
$\Delta\mathbf{p}^{\mathrm{T}}, \Delta\mathbf{p}^{\mathrm{B}}$	Tool-frame and base-frame representations of one displacement. Superscripts name frames.
$\mathbf{R}_{\mathrm{BT}}$	Rotation mapping tool-frame vectors into base coordinates.
$F_{\Delta t},\ \kappa$	Physical transition over interval Δt, and controller/environment settings. In Chapter 6, κ is specifically a scalar deadband width.
$\mathcal{O},\ \varepsilon_t^s,\ \varepsilon_t^o$	Observation map, process error, and observation error in the teaching state model.
$\mathbf{x}_t, \mathbf{v}_t, \mathbf{M}$	Cartesian positions, physical velocities, and a constant mass matrix in the simplified mechanics example.
$\mathbf{f}_{\text{int}}, \mathbf{f}_{\text{ext}}$	Internal and applied forces in those coordinates.
$\mathbf{J}_{\mathrm{c}}, \mathbf{f}_{\mathrm{c},t}$	Contact Jacobian and contact force in contact coordinates.
$\mathrm{gap}(\mathbf{x}),\ f_{\mathrm{n}}$	Normal surface separation and compressive normal contact force.
T_{cycle}	Elapsed time of one sequential prediction/decision cycle.
$T_{\text{policy}}, T_{\text{dynamics}}$	Policy-sampling and world-prediction elapsed times.
$T_{\text{decode}}, T_{\text{judge}}, T_{\text{transfer}}$	Decoding, scoring, and data-transfer elapsed times.

Operators and recurring abbreviations

Notation	Meaning
$\sim,\ \lvert,\ \cdot$	Sampled from; conditioned on; an unspecified argument of a distribution.
$\mathbb{R},\ \mathcal{N}(0, I)$	Real numbers and a zero-mean Gaussian with identity covariance.
$\mathbb{E},\ \nabla_\theta$	Expectation over specified random inputs and gradient with respect to θ.
$\lVert\cdot\rVert,\ (\cdot)^\top$	Euclidean norm (flattening an array when needed) and transpose.
sg	Stop-gradient: preserve the value while setting its derivative to zero for the update.
$\mathrm{clip}(x, l, u)$	Restrict scalar x to the interval with lower bound l and upper bound u.

Notation	Meaning
$\arg\max$	Return an argument attaining the maximum; use a specified tie rule.
$\mathrm{softmax}, \mathrm{Attn}$	Normalize scores into nonnegative weights summing to one; scaled dot-product attention.
append	Add a new action and observation to a history.
CFM, ODE, NFE	Conditional flow matching; ordinary differential equation; number of function evaluations.
RL, TD, GAE	Reinforcement learning; temporal difference; generalized advantage estimation.
PMPO	Preference-based maximum a posteriori policy optimization.
PPO, FPO, GRPO	Proximal Policy Optimization; Flow Policy Optimization; group-relative policy optimization.
ELBO, KL, MSE	Evidence lower bound; Kullback–Leibler divergence; mean squared error.
VLA, VLM	Vision–language–action model; vision–language model.
RGB, KV	red–green–blue image channels; attention keys and values.
FEM	Finite element method.
TP, FP, FN, TN	True positives, false positives, false negatives, and true negatives, relative to the stated success label.

The published model names π_0, $\pi_{0.5}$, and $\pi^*_{0.6}$ are names, not time indices or instances of the parameter subscript θ.

Task and context arguments may be suppressed after they are defined, but their meaning does not change. The same rule applies to batch and time indices. An equation using one action can be extended to a chunk only after specifying its joint density, executed duration, reward aggregation, and discount interval.